

EXERCISE 10: ONE-PHASE T-TEST

ANSC N/A

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REVIEWER SYMBOLS:

DLA-18
ARMY-AR
AIR FORCE-82

USER SYMBOLS:

DD FORM 672-1 (Continued)

Obsolete portion of this form has been left

P.A. - AS NAVY - AS Other Com USAF - 99 ARMY - AV	TITLE BOLT, TENSION, STEEL, 180 KSI FTU, 450° F. 12 POINT, EXTERNAL WRENCHING FLANGED	PROCURMENT SPECIFICATION MIL-B-8831	SUPERSEDES MS21250(ASG)	SHEET 2 OF 3 MILITARY STANDARD MS21250
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DIA DASH NO.	THREAD MIL-S- 8879	A DIA	B DIA	NOM SIZE W (EX)	D DIA	E MIN	H	J +.010 -.030	M	N DIA +.005	(a) P MAX	Q	R	(a) S MAX	(d) T MIN REF	U +.016	DRILL SIZE REF	CONCENTRICITY SEE (a), (b) & (c)			DOUBLE SHEAR LBS MIN	ULT TENSILE LBS MIN
																		X	Y	Z		
02	.1640-32 UNJC-3A	.300 .290	.1635 .1625	.2188	.219 .204	.076	.250	.100	.062	.037	-	.050	.041 .031	-	.385	.031	-	.004	.003	.003	4,600	2,950
03	.1900-32 UNJF-3A	.350 .340	.1895 .1885	.2500	.250 .235	.092	.265	.125			-	.055		-	.420		-	.004	.003	.004	6,100	3,910
04	.2500-28 UNJF-3A	.438 .428	.2495 .2485	.3125	.312 .297	.133	.300	.180			-	.069		-	.492		-	.005	.004	.0045	10,600	6,980
05	.3125-24 UNJF-3A	.531 .521	.3120 .3110	.3750	.375 .360	.162	.348	.215	.070	.055	-	.082	.057 .047	-	.579	.047	-	.006	.005	.005	16,600	11,100
06	.3750-24 UNJF-3A	.649 .639	.3745 .3735	.4375	.437 .422	.197	.388	.260			-	.091		-	.625		-	.008	.006	.006	23,900	17,100
07	.4375-20 UNJF-3A	.750 .740	.4370 .4360	.5000	.500 .485	.228	.435	.320			-	.099		-	.721		-	.009	.008	.008	32,500	23,200
08	.5000-20 UNJF-3A	.828 .818	.4995 .4985	.5625	.562 .547	.254	.504	.380	.094	.055	-	.123	.073 .063	-	.768	.062	-	.010	.008	.008	42,400	30,900
09	.5625-18 UNJF-3A	.938 .928	.5615 .5605	.6250	.625 .610	.287	.557	.440			-	.133		-	.852		-	.011	.009	.006	53,700	39,200
10	.6250-18 UNJF-3A	1.050 1.040	.6240 .6230	.6875	.687 .672	.327	.618	.500			-	.150		-	.899		-	.012	.010	.010	66,300	49,000
12	.7500-16 UNJF-3A	1.230 1.220	.7490 .7480	.8125	.812 .797	.380	.711	.570	.125	.055	.200	.178	.187 .163	-	1.036	.078	2	.015	.012	.012	95,400	71,100
14	.8750-14 UNJF-3A	1.438 1.428	.8740 .8730	.9375	.937 .922	.438	.808	.650			.260	.198		-	1.244		3	.018	.014	.014	129,900	97,100
16	1.0000-12 UNJF-3A	1.625 1.615	.9990 .9980	1.0625	1.062 1.047	.493	.923	.740			.260	.222	.250	-	1.479		3	.020	.016	.016	169,600	126,000
18	1.1250-12 UNJF-3A	1.875 1.865	1.1240 1.1225	1.2500	1.250 1.235	.556	1.051	.840	.188	.320	.258	.269		-	1.650	.094	4	.022	.019	.019	214,700	162,000
20	1.2500-12 UNJF-3A	2.125 2.115	1.2490 1.2475	1.3125	1.312 1.297	.636	1.155	.950			.274	.274	.089 .077	-	1.760		4	.023	.021	.021	265,100	202,000
22	1.3750-12 UNJF-3A	2.313 2.303	1.3740 1.3725	1.4375	1.437 1.422	.698	1.266	1.070			.335	.335		-	1.885		4	.028	.023	.023	320,700	247,000
24	1.5000-12 UNJF-3A	2.500 2.490	1.4990 1.4975	1.6250	1.625 1.610	.750	1.434	1.200						-	2.010			.030	.025	.025	381,700	296,000

- (a) DIMENSIONS A, B AND J SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR X FIM.
 (b) DIMENSIONS W AND J SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR Y FIM.
 (c) DIMENSIONS B AND THREAD PITCH DIAMETER SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR Z FIM.
 (d) REFERENCE DIMENSIONS ARE FOR DESIGN PURPOSES ONLY AND ARE NOT AN INSPECTION OR MANUFACTURING REQUIREMENT.
 (e) ON BOLTS 3/4 INCH OR LARGER, CENTER DRILLING OF SHANK IS OPTIONAL FOR MANUFACTURER.
 (f) ONLY BOLTS FOR WHICH THERE ARE QUALIFIED PRODUCTS LISTED IN QPL8831 SHALL BE USED FOR DESIGN.
 (g) WRENCHING ELEMENT SHALL BE IN ACCORDANCE WITH AS 870.

APPROVED 23 NOV 60 REVISED (L) FOR CHANGES SEE SHEET 1 AND 2

TABLE 1

FED. SUP CLASS
5306

TABLE 11

FED. SUP CLASS
5306

GRIP DASH NO.	GRIP # .010	DIAMETER DASH NUMBERS															
		02	03	04	05	06	07	08	09	10	12	14	16	18	20	22	24
LENGTH																	
004	.250	.655	.690	.762	.849	.895	.991	1.038									
006	.375	.780	.815	.887	.974	1.020	1.116	1.163	1.247	1.294	1.431						
008	.500	.905	.940	1.012	1.099	1.145	1.241	1.288	1.372	1.419	1.556	1.764	1.999	2.170			
010	.625	1.030	1.065	1.137	1.224	1.270	1.366	1.413	1.497	1.544	1.681	1.889	2.124	2.295	2.405	2.530	2.655
012	.750	1.155	1.190	1.262	1.349	1.395	1.491	1.538	1.622	1.669	1.806	2.014	2.249	2.420	2.530	2.655	2.780
014	.875	1.280	1.315	1.387	1.474	1.520	1.616	1.663	1.747	1.794	1.931	2.139	2.374	2.545	2.655	2.780	2.905
016	1.000	1.405	1.440	1.512	1.599	1.645	1.741	1.788	1.872	1.919	2.056	2.264	2.499	2.670	2.780	2.905	3.030
018	1.125	1.530	1.565	1.637	1.724	1.770	1.866	1.913	1.997	2.044	2.181	2.389	2.624	2.795	2.905	3.030	3.155
020	1.250	1.655	1.690	1.762	1.849	1.895	1.991	2.038	2.122	2.169	2.306	2.514	2.749	2.920	3.030	3.155	3.280
022	1.375	1.780	1.815	1.887	1.974	2.020	2.116	2.163	2.247	2.294	2.431	2.639	2.874	3.045	3.155	3.280	3.405
024	1.500	1.905	1.940	2.012	2.099	2.145	2.241	2.288	2.372	2.419	2.556	2.764	2.999	3.170	3.280	3.405	3.530
026	1.625	2.030	2.065	2.137	2.224	2.270	2.366	2.413	2.497	2.544	2.681	2.889	3.124	3.295	3.405	3.530	3.655
028	1.750	2.155	2.190	2.262	2.349	2.395	2.491	2.538	2.622	2.669	2.806	3.014	3.249	3.420	3.530	3.655	3.780
030	1.875	2.280	2.315	2.387	2.474	2.520	2.616	2.663	2.747	2.794	2.931	3.139	3.374	3.545	3.655	3.780	3.905
032	2.000	2.405	2.440	2.512	2.599	2.645	2.741	2.788	2.872	2.919	3.056	3.264	3.499	3.670	3.780	3.905	4.030
034	2.125	2.530	2.565	2.637	2.724	2.770	2.866	2.913	2.997	3.044	3.181	3.389	3.624	3.795	3.905	4.030	4.155
035	2.187					2.833											
036	2.250	2.655	2.690	2.762	2.849	2.895	2.991	3.038	3.122	3.169	3.306	3.514	3.749	3.920	4.030	4.155	4.280
038	2.375	2.780	2.815	2.887	2.974	3.020	3.116	3.163	3.247	3.294	3.431	3.639	3.874	4.045	4.155	4.280	4.405
039	2.437							3.225									
040	2.500	2.905	2.940	3.012	3.099	3.145	3.241	3.288	3.372	3.419	3.556	3.764	3.999	4.170	4.280	4.405	4.530
042	2.625	3.030	3.065	3.137	3.224	3.270	3.366	3.413	3.497	3.544	3.681	3.889	4.124	4.295	4.405	4.530	4.655
044	2.750	3.155	3.190	3.262	3.349	3.395	3.491	3.538	3.622	3.669	3.806	4.014	4.249	4.420	4.530	4.655	4.780
046	2.875	3.280	3.315	3.387	3.474	3.520	3.616	3.663	3.747	3.794	3.931	4.139	4.374	4.545	4.655	4.780	4.905
048	3.000	3.405	3.440	3.512	3.599	3.645	3.741	3.788	3.872	3.919	4.056	4.264	4.499	4.670	4.780	4.905	5.030
050	3.125	3.530	3.565	3.637	3.724	3.770	3.866	3.913	3.997	4.044	4.181	4.389	4.624	4.795	4.905	5.030	5.155
052	3.250	3.655	3.690	3.762	3.849	3.895	3.991	4.038	4.122	4.169	4.306	4.514	4.749	4.920	5.030	5.155	5.280
054	3.375	3.780	3.815	3.887	3.974	4.020	4.116	4.163	4.247	4.294	4.431	4.639	4.874	5.045	5.155	5.280	5.405
056	3.500	3.905	3.940	4.012	4.099	4.145	4.241	4.288	4.372	4.419	4.556	4.764	4.999	5.170	5.280	5.405	5.530
058	3.625	4.030	4.065	4.137	4.224	4.270	4.366	4.413	4.497	4.544	4.681	4.889	5.124	5.295	5.405	5.530	5.655
060	3.750	4.155	4.190	4.262	4.349	4.395	4.491	4.538	4.622	4.669	4.806	5.014	5.249	5.420	5.530	5.655	5.780
062	3.875	4.280	4.315	4.387	4.474	4.520	4.616	4.663	4.747	4.794	4.931	5.139	5.374	5.545	5.655	5.780	5.905
064	4.000	4.405	4.440	4.512	4.599	4.645	4.741	4.788	4.872	4.919	5.056	5.264	5.499	5.670	5.780	5.905	6.030
066	4.125	4.530	4.565	4.637	4.724	4.770	4.866	4.913	4.997	5.044	5.181	5.389	5.624	5.795	5.905	6.030	6.155
068	4.250	4.655	4.690	4.762	4.849	4.895	4.991	5.038	5.122	5.169	5.306	5.514	5.749	5.920	6.030	6.155	6.280
070	4.375	4.780	4.815	4.887	4.974	5.020	5.116	5.163	5.247	5.294	5.431	5.639	5.874	6.045	6.155	6.280	6.405
072	4.500	4.905	4.940	5.012	5.099	5.145	5.241	5.288	5.372	5.419	5.556	5.764	5.999	6.170	6.280	6.405	6.530
074	4.625	5.030	5.065	5.137	5.224	5.270	5.366	5.413	5.497	5.544	5.681	5.889	6.124	6.295	6.405	6.530	6.655
076	4.750	5.155	5.190	5.262	5.349	5.395	5.491	5.538	5.622	5.669	5.806	6.014	6.249	6.420	6.530	6.655	6.780
078	4.875	5.280	5.315	5.387	5.474	5.520	5.616	5.663	5.747	5.794	5.931	6.139	6.374	6.545	6.655	6.780	6.905
080	5.000	5.405	5.440	5.512	5.599	5.645	5.741	5.788	5.872	5.919	6.056	6.264	6.499	6.670	6.780	6.905	7.030
082	5.125	5.530	5.565	5.637	5.724	5.770	5.866	5.913	5.997	6.044	6.181	6.389	6.624	6.795	6.905	7.030	7.155
084	5.250	5.655	5.690	5.762	5.849	5.895	5.991	6.038	6.122	6.169	6.306	6.514	6.749	6.920	7.030	7.155	7.280
086	5.375	5.780	5.815	5.887	5.974	6.020	6.116	6.163	6.247	6.294	6.431	6.639	6.874	7.045	7.155	7.280	7.405
088	5.500	5.905	5.940	6.012	6.099	6.145	6.241	6.288	6.372	6.419	6.556	6.764	6.999	7.170	7.280	7.405	7.530
090	5.625	6.030	6.065	6.137	6.224	6.270	6.366	6.413	6.497	6.544	6.681	6.889	7.124	7.295	7.405	7.530	7.655
092	5.750	6.155	6.190	6.262	6.349	6.395	6.491	6.538	6.622	6.669	6.806	7.014	7.249	7.420	7.530	7.655	7.780
094	5.875	6.280	6.315	6.387	6.474	6.520	6.616	6.663	6.747	6.794	6.931	7.139	7.374	7.545	7.655	7.780	7.905
096	6.000	6.405	6.440	6.512	6.599	6.645	6.741	6.788	6.872	6.919	7.056	7.264	7.499	7.670	7.780	7.905	8.030
098	6.125								6.997	7.044	7.181	7.389	7.624	7.795	7.905	8.030	8.155
100	6.250								7.122	7.169	7.306	7.514	7.749	7.920	8.030	8.155	8.280
102	6.375								7.247	7.294	7.431	7.639	7.874	8.045	8.155	8.280	8.405
104	6.500								7.372	7.419	7.556	7.764	7.999	8.170	8.280	8.405	8.530
106	6.625								7.497	7.544	7.681	7.889	8.124	8.295	8.405	8.530	8.655
108	6.750								7.622	7.669	7.806	8.014	8.249	8.420	8.530	8.655	8.780
110	6.875								7.747	7.794	7.931	8.139	8.374	8.545	8.655	8.780	8.905
112	7.000								7.872	7.919	8.056	8.264	8.499	8.670	8.780	8.905	9.030
114	7.125													8.795	8.905	9.030	9.155
116	7.250													8.920	9.030	9.155	9.280
118	7.375													9.045	9.155	9.280	9.405
120	7.500													9.170	9.280	9.405	9.530
122	7.625													9.295	9.405	9.530	9.655
124	7.750													9.420	9.530	9.655	9.780
126	7.875													9.545	9.655	9.780	9.905
128	8.000													9.670	9.780	9.905	10.030

USER SYMBOLS:
DLA-18
ARMY-AR
AIR FORCE-82

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USAF - 99
ARMY - AV

PROCUREMENT SPECIFICATION
MIL-B-8831

TITLE

BOLT, TENSION, STEEL, 180 KSI FTU, 450°F,
12 POINT, EXTERNAL WRENCHING FLANGED

SUPERSEDES:

MS21250(ASG)

MILITARY STANDARD

MS21250

SHEET 3 OF 3

DD FORM 672-1 (Coordinated)

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